Practical 3
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Batch: A3

```
import time

import os

from datetime import datetime


# Create output folders

os.makedirs("captured_images", exist_ok=True)

os.makedirs("captured_videos", exist_ok=True)


def capture_image():

    filename =
f"captured_images/image_{datetime.now().strftime('%Y%m%d_%H%M%S')}.jpg"

    # Simulate image capture by creating a dummy file with metadata

    with open(filename, "w") as f:

        f.write(f"Simulated Image\nCaptured at: {datetime.now()}\nResolution:
1920x1080\n")

    return filename


def record_video(duration=3):

    filename =
f"captured_videos/video_{datetime.now().strftime('%Y%m%d_%H%M%S')}.h264"

    # Simulate video recording

    with open(filename, "w") as f:

        f.write(f"Simulated Video\nRecorded at: {datetime.now()}\nDuration:
{duration}s\nFormat: H264\n")

    return filename
```

```python
print("=" * 40)
print("      Pi Camera Simulation")
print("=" * 40)

# ── Image Capture ──────────────────────────────
print("\n[LED PIN 40 ON] Camera is ON...")
time.sleep(0.5)

print("Capturing image", end="", flush=True)
for _ in range(3):
    time.sleep(0.4)
    print(".", end="", flush=True)

img_file = capture_image()
print(f"\n✓ Image Captured — saved as: {img_file}")
print("[LED PIN 40 OFF]")

time.sleep(1)

# ── Video Recording ────────────────────────────
print("\n[LED PIN 38 ON] Starting video recording...")
print("Recording video", end="", flush=True)
for i in range(3):
    time.sleep(1)
    print(f" {i+1}s", end="", flush=True)

vid_file = record_video(duration=3)
```

```
print(f"\n✓ Video Recorded — saved as: {vid_file}")
print("[LED PIN 38 OFF]")

print("\n" + "=" * 40)
print("Done! Files saved:")
print(f"  📷 {img_file}")
print(f"  🎥 {vid_file}")
print("=" * 40)
```

Output

```
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    Pi Camera Simulation
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[LED PIN 40 ON] Camera is ON...
Capturing image...
✓ Image Captured — saved as: captured_images/image_20260509_165821.jpg
[LED PIN 40 OFF]

[LED PIN 38 ON] Starting video recording...
Recording video 1s 2s 3s
✓ Video Recorded — saved as: captured_videos/video_20260509_165825.h264
[LED PIN 38 OFF]

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Done! Files saved:
  📷 captured_images/image_20260509_165821.jpg
```

🎥 **captured_videos/video_20260509_165825.h264**

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